

Markscheme

Mathematics: analysis and approaches
Practice question bank

175. (a)

Consider the system $x + y + z = 6$, $2x - y + z = 3$, $x + 4y + kz = 15$.

Find the value of k for which this system does not have a unique solution.

a unique solution fails when the determinant of the coefficient matrix is zero (M1)

$$\det = 1(-k - 4) - 1(2k - 1) + 1(8 + 1) \quad (M1) \text{ A1}$$

$$= -3k + 6 \quad \text{A1}$$

$$\text{set } -3k + 6 = 0: k = 2 \quad \text{A1}$$

[7 marks]

(b)

Hence, using a value of k for which the system does have a unique solution, find x , y and z when $k = 0$.

$$\text{with } k = 0: (3) \text{ becomes } x + 4y = 15; \text{ subtract (2) from (1): } -x + 2y = 3 \quad (M1)$$

$$\text{solving simultaneously: } y = 3, x = 3 \quad (M1) \text{ A1}$$

$$\text{from (1): } z = 6 - 3 - 3 = 0 \quad \text{A1}$$

[3 marks]